

Bala Harshith S

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AI Engineer | LLM Applications, AI Agents, RAG

PROFESSIONAL SUMMARY

AI Engineer with 4+ years of experience designing, deploying, and operating production AI/ML solutions, specializing in Large Language Models (LLMs), Generative AI, intelligent agents, and Retrieval-Augmented Generation (RAG). Hands-on expertise in Python, LangChain, LlamaIndex, LangGraph, CrewAI, OpenAI and Azure OpenAI APIs, open-source model fine-tuning, vector search, FastAPI microservices, evaluation, guardrails, and cloud-native deployment. Builds scalable AI applications optimized for accuracy, latency, reliability, security, and operational cost.

TECHNICAL SKILLS

LLMs & Generative AI: Large Language Models, OpenAI APIs, Azure OpenAI, LLaMA, Hugging Face Transformers, PEFT, LoRA, QLoRA, Prompt Optimization

Agents & Orchestration: LangChain, LlamaIndex, LangGraph, CrewAI, MCP, Tool Calling, Planning, Memory, Context Management, Structured Outputs

RAG & Vector Search: Document Ingestion, Chunking, Embeddings, Semantic Search, Hybrid Search, Reranking, Pinecone, FAISS, pgvector, Azure AI Search

Programming & Backend: Python, SQL, FastAPI, Flask, REST APIs, Microservices, JSON, Pydantic, PostgreSQL, Async Workflows, Git

Cloud, DevOps & MLOps: AWS, Azure, SageMaker, Azure Functions, Docker, Kubernetes, MLflow, GitHub Actions, CI/CD, Model Monitoring

Evaluation, Safety & Performance: Retrieval Precision, Groundedness, Hallucination Rate, Schema Validity, Content Moderation, Guardrails, Latency, Token Cost, Regression Testing, Observability

PROFESSIONAL EXPERIENCE

McKesson | Gen AI Engineer | Aug 2024 - Present

- Designed and deployed production LLM-powered applications using Python, FastAPI, Azure OpenAI, LangChain, LlamaIndex, LangGraph, Azure AI Search, and cloud-native microservices.
- Built intelligent agents and multi-agent workflows with planning, tool calling, memory, context management, structured outputs, retries, fallbacks, and reusable orchestration components.
- Engineered RAG pipelines across document ingestion, chunking, embeddings, vector indexing, semantic and hybrid search, metadata filtering, reranking, grounding, citations, and retrieval optimization.
- Fine-tuned open-source LLMs with Hugging Face PEFT, LoRA, and QLoRA and optimized prompts, inference latency, token usage, model quality, and serving cost for production workloads.
- Implemented guardrails, content moderation, access controls, and evaluation workflows for retrieval precision, groundedness, hallucination rate, schema validity, latency, regressions, and production monitoring.

PNC Financial | AI/ML Engineer | Dec 2022 - Nov 2023

- Built production ML and NLP solutions using Python, SQL, Hugging Face Transformers, PyTorch, TensorFlow, Spark, AWS SageMaker, and large-scale financial and regulatory datasets.
- Developed Transformer-based document understanding, text classification, semantic retrieval, and structured information extraction workflows using open-source models and embedding-based search.
- Created FastAPI and Flask REST services integrating model inference with relational databases, internal APIs, reporting systems, and downstream financial applications.
- Evaluated classification, regression, and anomaly-detection models across more than 12 million transactions, balancing accuracy, explainability, latency, reliability, and operational cost.
- Automated model deployment, monitoring, and regression testing with SageMaker, Lambda, CloudWatch, Docker, and CI/CD, improving deployment consistency by 30%.

Macy's | Data Scientist | Jun 2021 - Nov 2022

- Built recommendation, personalization, forecasting, classification, and clustering models using Python, SQL, Scikit-learn, XGBoost, collaborative filtering, and customer behavior features.
- Developed PySpark, Databricks, SQL, pandas, and NumPy workflows supporting model development, feature processing, experimentation, and customer-facing digital commerce use cases.
- Evaluated model and business performance through offline metrics, KPI dashboards, and stakeholder feedback, improving targeting accuracy by 25%.

PROJECTS

Production Agentic RAG & LLM Platform | Python, FastAPI, LangGraph, LlamaIndex, Pinecone/pgvector, Docker, Kubernetes

- Built a production-shaped agentic platform connecting enterprise documents, RAG retrieval, SQL data, vector databases, API tools, agent memory, multi-step workflows, evaluation, and user-facing responses.
- Implemented LoRA/QLoRA adapters, prompt libraries, guardrails, content filters, structured outputs, retrieval and LLM evaluations, Docker/Kubernetes deployment assets, monitoring, and GitHub CI/CD checks.

CERTIFICATIONS

- Oracle Cloud AI Infrastructure 2025

EDUCATION

Master of Science in Computer Science - University of Central Missouri

Bachelor of Technology in Electronics and Communication Engineering - Teegala Krishna Reddy Engineering College